HARSHIT SINGH SHEKHAWAT

Email: harshitshekhawa501@gmail.com LinkedIn: linkedin.com/in/harshitshekhawat GitHub: github.com/lawapaul
Phone: +91 7340532228 Codolio : https://codolio.com/profile/surferw96

EDUCATION

Vellore Institute of Technology
Aug 2023 - Present

Bachelor of Technology
CGPA: 8.95/10

Major in Computer Science, Specialization in Artificial Intelligence & Machine Learning
Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, AWS Cloud Computing, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Computer Architecture

Delhi Public School Jodhpur
Feb 2009 - May 2023

Completed secondary and senior secondary education under the CBSE board.

SKILLS

- Programming: Python, C++, Java, JavaScript, SQL, C
- AI/ML: Machine Learning, Deep Learning, Neural Networks, NLP, Computer Vision, Scikit-learn, TensorFlow, PyTorch
- Data & Analytics: Pandas, NumPy, SQL, Data Preprocessing, Model Evaluation
- Cloud & Tools: AWS, Docker, Git, Linux, MATLAB, R Studio
- IoT & Embedded: Arduino, ESP32, IoT Sensors
- Databases: MongoDB, MySQL, SQLite

PROJECTS

Agentic AI Intrusion Detection & Response Pipeline | TensorFlow, PyTorch, FastAPI
- Engineered an explainable AI-driven IDS using CNN/LSTM/GRU/Transformer models on ~2.8M CICIDS2017 records with graph-based attack correlation, LLM reasoning, and adaptive risk fusion.
- Achieved 98.40% accuracy and 99.16% memory retrieval accuracy with human-in-the-loop feedback and retraining workflows.

ParkVision – Parking Guidance & Number Plate Recognition System | OpenCV, YOLOv8, FastAPI, React
- Built a computer vision parking system using YOLOv8, EasyOCR, and scikit-learn for real-time occupancy detection and automatic number plate recognition across 396 parking slots.
- Developed a FastAPI + React dashboard with live parking analytics, slot recommendation, duplicate suppression, and SQLite-based vehicle tracking.

Explainable Plant Disease Detection Using Deep Learning | TensorFlow, RAG, LLMs, HITL
- Developed a deep learning pipeline to classify four apple leaf categories—Healthy, Scab, Black Rot, and Cedar Apple Rust—using 7,771 images from the PlantVillage dataset.
- Integrated explainable AI techniques with a Retrieval-Augmented Generation (RAG) pipeline and Large Language Models (LLMs) to provide context-aware interpretations of disease predictions and retrieve relevant disease information.
- Designed a human-in-the-loop feedback mechanism to capture user corrections, maintain feedback memory, and support continuous improvement of the explainable disease detection system.

Smart Irrigation System (AquaField) | Flask, Scikit-learn, ESP32, Arduino
- Co-developed an ML-powered irrigation system using Flask, ESP32, and IoT sensors, integrating real-time monitoring, rain-aware motor control, and hardware safety cutoffs.
- Benchmarked 4 regression models on 100K agricultural records, achieving an R^2 of 0.99870 with Gradient Boosting.
- Analyzed real-time soil moisture, rainfall, temperature, and water-flow data for efficient irrigation scheduling.

VOLUNTEER EXPERIENCE

General Secretary – Mharo Rajasthan Club **September 2024 - April 2026**
Directed overall club operations by coordinating cultural and technical events, increasing the number of club-conducted events by 40% and helping regain club popularity and student engagement. Led the planning and execution of a large-scale event for 300+ students with overall collections nearing ₹1 lakh while managing coordination, budgeting, and operational workflows.

Management Team Member – Linux Club **June 2024 - April 2025**
Organized workshops and hackathons focused on Linux, open-source tools, and system-level programming concepts. Managed event planning, scheduling, and team coordination to ensure smooth execution of technical sessions and promote hands-on learning within the student community.

Photographer – Photography Club **July 2024 - May 2025**
Planned and covered photography events and exhibitions while contributing to creative promotion and visual storytelling activities. Collaborated with club members to improve event outreach, audience engagement, and participation through curated media content.